MATH 152 **EXAM II** NAME _____

Mrs. Bonny Tighe 50 points

7.7-11.7 TA_____ Wed 10/30/2019

PART II

1. (10 points) Complete the following tests using no abbreviations.

GEOMETRIC SERIES $\sum_{n=0}^{\infty} a_n (r)^n$ is _____ if $|r| \geq 1$

THE COMPARISON TEST Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

i) If $\sum_{n=1}^{\infty} b_n$ is convergent and a_n ____ b_n for all n, then $\sum_{n=1}^{\infty} a_n$ is also convergent

ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and a_n ____ b_n for all n, then $\sum_{n=1}^{\infty} a_n$ is also divergent

THE LIMIT COMPARISON TEST

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms . If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is _____, then either both series converge or both diverge.

Name the test:_____ If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a _____, _____ and _____function on [1, ∞) and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if _____is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is _____ if $p > 1$

RATIO TEST

If $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L$ $\begin{cases} L < 1 \text{ then } \sum a_n \text{ is } _____ \\ L > 1 \text{ then } \sum a_n \text{ is } _____ \\ L = 1 \text{ then the test is } _____ \end{cases}$

ROOT TEST

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ and $\begin{cases} L < 1 \text{ then } \sum a_n \text{ is } _____ \\ L > 1 \text{ then } \sum a_n \text{ is } _____ \\ L = 1 \text{ then the root test is } _____ \end{cases}$

THE ALTERNATING SERIES TEST

If i) if $b_{n+1} \leq b_n$ for all n and ii) $\lim_{n\to\infty} b_n \neq 0$ Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is _____

REMAINDER ESTIMATE FOR THE INTEGRAL TEST

Suppose $a_n = f(n)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent . If $R_n = S - S_n$, then _____ $\leq R_n \leq$ _____

2. (5 points). Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=0}^{\infty} \frac{e^n}{e^{2n}+1}$$

3. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $$\sum_{n=1}^{\infty} \frac{1}{n^2+3n+2}$$

b) $$\sum_{n=1}^{\infty} \frac{\pi^{n+1}}{2^{2n}}$$

4 (10 points) Test the following series for absolute convergence, conditional convergence or divergence. State the test you are using and show all your work.

a) $\sum_{n=1}^{\infty} \frac{1}{n+5^n}$

b) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+2n+1}}$

c) $\sum_{n=1}^{\infty} \frac{(3 \cdot 5 \cdot 7 \cdot 9 \cdot (2n+1))}{3^n \, n!}$

5. (10 points) The given curve is rotated about the x-axis. Find the area of the resulting surface. $S = 2\pi \int radius\ (arclength)$ $\quad x = \ln(y^2 - 1), \quad 2 \le y \le 3$

6. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for Simpson's rule. Suppose $|f''''(x)| \le K \quad for \quad a \le x \le b$, then

$$|E_S| \le \frac{K(b-a)^5}{180n^4} \qquad \int_1^2 x\sqrt[3]{x}\,dx, \quad n = 4$$

How large should we take *n* in order to guarantee that the Simpson's rule approximation for $\int_1^2 x\sqrt[3]{x}\,dx,$ is accurate to within 0.01?

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PART I

1. (12 @ 3 pts each) Multiple Choice: Circle the correct letter

a) Find a sequence for $1, \frac{1 \cdot 5}{1}, \frac{1 \cdot 5 \cdot 9}{1 \cdot 2}, \frac{1 \cdot 5 \cdot 9 \cdot 13}{1 \cdot 2 \cdot 3}, \ldots$ is

a) $a_n = \frac{4n - 3}{n!}$

b) $a_n = \frac{(1 \cdot 5 \cdot 9 \cdots (4n - 3))}{(n - 1)!}$

c) $a_n = \frac{(1 \cdot 5 \cdot 9 \cdots (4n - 3))}{n!}$

d) $a_n = \frac{4n - 1}{(n - 1)!}$

e) None of the above

b) $\sum_{n=1}^{\infty} \frac{(-5)^{n+1}}{3^{2n}}$ equals

a) $\frac{-45}{14}$

b) $\frac{25}{14}$

c) $\frac{25}{4}$

d) $\frac{-45}{4}$

e) None of the above

c) Which test should you use to test for absolute convergence of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n^3 + 4n + 2}}$

a) Alternating Series Test
b) Integral Test
c) Limit Comparison Test
d) Ratio Test
e) None of the above tests

d) Which of the following sequences is not convergent?

a) $a_n = \frac{5^n + 2}{3^n}$

b) $a_n = \frac{4n-1}{2n+1}$

c) $a_n = \frac{3}{n!}$

d) $a_n = \ln(n+2) - \ln(n+3)$

e) All the sequences are convergent

f) More than one sequence is divergent

e)For which of the following can we NOT use the Integral Test to determine convergence or divergence?

a) $\sum_{n=0}^{\infty} \frac{(-1)^n}{1+n^2}$

b) $\sum_{n=1}^{\infty} \frac{3^{n+2}}{4^{n-1}}$

c) $\sum_{n=0}^{\infty} \frac{n}{n^2+4}$

d) $\sum_{n=1}^{\infty} n^3 e^{-n^4}$

e) The Integral test can be used for all the above series.

f)Which of the following inequalities are needed to show the convergence of $\int_1^{\infty} \frac{x}{\sqrt{x^6+3x^2+2}} dx$

a) $\frac{x}{\sqrt{x^6+3x^2+2}} \geq \frac{1}{x^2}$

b) $\frac{x}{\sqrt{x^6+3x^2+2}} \geq \frac{1}{x^3}$

c) $\frac{x}{\sqrt{x^6+3x^2+2}} \leq \frac{1}{x^2}$

d) $\frac{x}{\sqrt{x^6+3x^2+2}} \leq \frac{1}{x^3}$

e) None of the above

g)Using the Alternating Series Estimation Theorem we can approximate the error when using S_8 of $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$ as

a) $R_8 \leq \frac{-1}{\sqrt{10}}$

b) $R_8 \leq \frac{1}{\sqrt{9}}$

c) $R_8 \leq \int_8^{\infty} \frac{(-1)^n}{\sqrt{n+1}} dn = 2\sqrt{9}$

d) $R_8 \leq \frac{1}{\sqrt{10}}$

e) The series does not converge, so we cannot estimate the error.

f) None of the above answers

h)Which test could you NOT use to determine the convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{e^n}$

a) Limit Comparison Test
b) Root Test
c) Integral Test
d) Ratio Test
e) You can use all of the above tests

i) The surface area for the function $x = e^y$ when rotated about the y-axis on the interval $2 \leq y \leq 3$ is:

a) $2\pi \int_2^3 y\sqrt{1+e^{2y}}\, dy$

b) $2\pi \int_2^3 e^y \sqrt{1+e^{2y}}\, dy$

c) $2\pi \int_2^3 x \sqrt{1+\left(\frac{1}{x^2}\right)}\, dx$

d) $2\pi \int_2^3 \ln x \sqrt{1+\left(\frac{1}{x^2}\right)}\, dx$

e) None of the above

j) The Error Bound formula for the Midpoint approximation is $|E_m| \leq \frac{K(b-a)^3}{24n^2}$,where $|f''(x)| \leq K \quad for \quad a \leq x \leq b$. Find K for approximating $\int_1^4 \sqrt{x}\, dx$ with 4 subdivisions.

a) $K = -\frac{1}{4}$

b) $K = \frac{1}{32}$

c) $K = \frac{1}{4}$

d) $K = \frac{1}{2}$

e) None of the above

k) Which of the following is **not** an improper integral?

a) $\int_{-2}^{5} \frac{1}{\sqrt{x+2}} dx$

b) $\int_{1}^{3} \frac{x}{x^2 + 3x - 4} dx$

c) $\int_{1}^{3} \frac{x}{x^2 - 3x + 4} dx$

d) $\int_{0}^{1} \frac{1}{\ln x} dx$

e) They are all improper integrals

l) Represent $0.\overline{746}$ as a series

a) $\sum_{n=0}^{\infty} \left(\frac{746}{1000}\right)^n$

b) $\sum_{n=1}^{\infty} \frac{746}{1000}\left(\frac{1}{1000}\right)^n$

c) $\sum_{n=0}^{\infty} 746\left(\frac{1}{1000}\right)^n$

d) $\sum_{n=0}^{\infty} \frac{746}{1000}\left(\frac{1}{1000}\right)^n$

e) None of the above

3. (19 @ 1 point each) Circle True or False

TRUE FALSE a) The first test you try when determining the convergence of a series is the limit comparison test.

TRUE FALSE b) Given $a_n = n^2 + 2n + 1$, then $a_{n+1} = n^2 + 4n + 4$.

TRUE FALSE c) To determine the convergence of $\sum_{n=0}^{\infty} \frac{n}{n^2 + n^3 + 1}$ I could use the Integral Test.

TRUE FALSE d) If $\lim_{n \to \infty} a_n = 0$, then the series $\sum_{n=0}^{\infty} a_n$ is convergent.

TRUE FALSE e) If $\lim_{n \to \infty} a_n = 0$, then the sequence $\{a_n\}$ is convergent.

TRUE FALSE f) The formula for the Trapezoid rule for approximation of a definite integral with four subdivisions is $\Delta x\left[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\right]$.

TRUE FALSE g) We would compare $\int_1^{\infty} \frac{x}{\sqrt{x^4 + 2x + 3}} dx$ to $\int_1^{\infty} \frac{1}{x^2} dx$ to determine convergence.

TRUE FALSE h) The arc length of the function $y = \sin x$ on the interval $0 \le y \le \frac{1}{2}$ is $\int_0^{\frac{1}{2}} \sqrt{1 + \left(\frac{1}{1 - y^2}\right)} dy$.

TRUE FALSE i) $\sum_{n=2}^{\infty} \frac{(-1)^n n}{\ln(n)}$ is conditionally convergent .

TRUE FALSE j) The Surface Area for the function $y = e^{2x}$ on the interval $0 \le x \le 1$ revolved about the x-axis is $2\pi \int_0^1 e^{2x} \sqrt{1 + 4e^{4x}}\, dx$

TRUE FALSE k) A series is convergent if and only if the sum exists.

TRUE FALSE l) $\int_0^4 \left(\frac{x}{x^2 - 1}\right) dx$ is an improper integral.

TRUE FALSE m) The trapezoid rule is more accurate than the midpoint rule when approximating the area under a curve.

TRUE FALSE n) The repeating decimal $0.6\overline{54}$ is equal to the series $\sum_{n=1}^{\infty} 0.654\left(\frac{1}{100}\right)^n$

TRUE FALSE o) $\int_1^{\infty}\left(\frac{3}{x^{\sqrt{2}}}\right)dx$ is absolutely convergent.

TRUE FALSE p) If f is continuous on $[0, \infty)$ and $\int_1^{\infty} f(x)\,dx$ is convergent, then $\int_0^{\infty} f(x)\,dx$ is also convergent

TRUE FALSE q) The Ratio Test can be used to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^5 + 3n^3 + 2n + 6}$ converges

TRUE FALSE r) If $0 \le a_n \le b_n$ and $\Sigma\, b_n$ diverges, then $\Sigma\, a_n$ diverges.

TRUE FALSE s) The root test proves that the series $\sum_{n=1}^{\infty}\left(\frac{4+2n}{5+2n}\right)^n$ is convergent.